LAST UPDATE: JULY 2022

104018	DIFFERENTIAL AND INTEGRAL CALCULUS 1M			
	Lectures 4	Tutorials 2	Lab/Proj	CREDITS 5
Prerequisites	None			

SYLLABUS

- The real numbers, sequences of real numbers.
- Real functions of one variable: limits and continuity, continuity on closed intervals, monotonicity and invertible functions.
- Differentiability, fundamental theorems of differential calculus, L'hospital's rule, Taylor's expansion, function investigation.
- Primitive function, indefinite integral and techniques of integrations.
- Definite integral and its properties, integrable functions, fundamental theorems of integral calculus.
- Improper integrals.
- Numerical series.
- Sequences of functions, series of functions and power series including radius of convergence.

REFERENCE BIBLIOGRAPHY

- "Thomas' Calculus", 14^{th} Edition by George B. Thomas.
- "Calculus and Analytic Geometry" by George B. Thomas.
- "Vector Calculus" by Jerrold E. Marsden.

REMARKS

This course is almost the same as Differential and Integral Calculus 1. The syllabus is the same, the lectures and tutorials can be delivered with the same level of depth and complexity. The most important difference is in the exams. As an example: even though in the lectures the notion of limit is presented in the fully formal manner with the ε-δ language, that is NOT included in the exams from 104003 Differential and Integral Calculus 1. The evaluation is only about technical skill when computing limits, not proving theorems or claims using the formal definition.

Here, in 104018 Differential and Integral Calculus 1M, it is a common practice to include some problems requiring the students to use some of the formal language from the proof of the theorems to solve the problems. Sometimes it is required in the exam to actually state and prove a given theorem.